Jump-starting AI agents

A practical quick start for economists

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About these slides

- **Objective:** get you started on one of your own projects right after skimming these slides.
- **For whom:** economists in academia, but much of this applies more widely.
- **Prerequisites:** almost none. To start on your project right away, you need a basic subscription (about 20 USD/month) with any AI provider.
- **Caveat:** this is my personal take in a field that changes very fast. It is what works for me, not a fixed recipe: adjust it to your own workflow and software.
- **Meta:** these slides were made with an agent, following exactly the workflow they describe.



Happy to discuss with anyone, any time: simon.martin@univie.ac.at

Chatbot vs agent

	Chatbot	Agent
Input	you copy/paste text into the chat	reads your files directly
Output	you copy/paste answers back into your documents	writes into your files (text, code, tables)
Context	the current chat only	your whole project folder
Tasks	answers one question at a time	plans and carries out multi-step tasks on its own

The basics on these slides get you maybe 90–95% of the gains. Ignore advanced features (skills, GitHub integration, . . .) for now: start simple.

“Coding agent”: not just for coding

- **Misconception 1: coding is for programmers.** Almost everything you do on a computer is “coding”: writing and formatting text, reading documents, calculating, combining content from several files.
- **Misconception 2: it is for empirical work only.** Agents work just as well for writing, admin and theory; their maths skills have improved a lot recently.

Writing

Check that the introduction of paper.tex is consistent with Section 5.

Admin

Draft the progress report from the documents in funding/.

Theory

Turn my notes in notes.md into a formal proposition, in the style of the samples in samples/.

Empirics

Add clustered standard errors to all regressions in code/analysis.do.

What it is, and what it isn't

"Just a token-guessing machine?"

- Technically, yes: it predicts the next piece of text.
- But predicting well at this scale means it can follow arguments, plan tasks and check proofs.
- Judge it by its results on your own tasks.

Not a store of truth

- It is not a database, and not Wikipedia.
- It reconstructs facts rather than looking them up, so they can be wrong, outdated or invented (citations!).

Strong at working with what you give it, weak at recalling facts. So provide the sources (literature/, data/), and let it compute in code instead of answering from memory (see slides 12 and 16).

What you need

Getting started

- **Subscription:** any basic plan at about 20 USD/month (Claude, ChatGPT/Codex, Gemini, . . .). Free plans are usually too limited; 200 USD plans are rarely needed. The provider matters less than you might think.
- **Software:** an editor with the agent built in, e.g. VS Code with the Claude Code extension (my setup).
- **Setup:** ask the agent to install or explain LaTeX, Stata paths etc., and to convert existing .docx/.pdf drafts to .tex/.md once.

Settings

- **Model:** any current model is fine. No need to chase every new release.
- **Effort:** medium for most tasks; it trades depth against speed and tokens. High only for occasional thorough checks.
- **Tokens:** the 20 USD plan suffices for most daily use. What eats tokens: many parallel web searches, and binary files (.docx, .xlsx, .pdf) instead of plain text.

Your agent is your research assistant

- Extremely capable, but only as good as the context you give it.
- Your job, as with any new RA:
  - **Context:** background, related literature, your terminology. A brilliant RA with a physics background needs to know how you use certain terms.
  - **Instructions:** what to do, which output you expect, and in which file.
  - **Unclear points:** say how to handle them, for example:

💬 *Never make assumptions. Always clarify with me first.*

💬 *Proceed as far as you can. Leave a TODO note in the document for anything unclear, so we can come back to it later.*

- You do most of this once, in `instructions.md` (see slide 13).

You stay in charge

- RAs make mistakes and misread input, and so does your agent. **You remain responsible for the final output.**
- **Verify:** read and check everything, and leave enough time for it at the end. AI slop shows.
- A second agent can cross-check for you, but the responsibility stays with you.
- **Memory drifts:** as a session grows, older or partly contradictory instructions get lost. Don't assume everything you ever said is still respected; now and then ask it to re-read the instructions or double-check.
- **Fresh sessions:** start new sessions rather than one endless chat. Let the agent record progress and open points in `status.md`, and read it at the start of each session.

Project = folder

- Agents work in **projects**, and a project is simply a folder on your computer.
- Everything the agent needs goes in there, and you come back to the same folder each time. Many projects are short-lived; that is fine.

A research paper

my-paper/
- instructions.md — the contract
- status.md — progress log
- paper.tex — main document
- literature/ — PDFs you may cite
- samples/ — your writing samples
- data/ — raw data
- code/ — do files
- output/ — generated tables, figures

These slides

jumpstart-slides/
- instructions.md — the contract
- status.md — progress log
- background.md — my notes
- recordings/ — voice transcripts
- structure.md — agreed outline
- examples/ — code shown on slides
- slides.tex — main document
- slides.pdf — final output

Plain text for everything you work on

Plain text	Binary
.tex, .md, .txt, .csv, .do, .R, .py	.docx, .xlsx, .pdf, .dta, .png
→ everything you actively work on	→ only pure input (literature, external files) and final output (the compiled PDF)

- **Why:** plain text is close to how the agent reads: much faster, far fewer tokens, and much easier to edit.
- **Example:** to find GDP for Austria in 2020, the agent first has to decode an .xlsx file. In a .csv, one quick search returns the number.
- **Bonus:** changes over time are easy to see (what changed, when, and why), even with simple Dropbox file history.

Try it: open a .docx or .pdf in Notepad or TextEdit. That is what the agent has to work through.

No copy/paste

- Copy/paste is friction: you should never need it.
- **Input:** point the agent to where things are.

> *Consider Smith2023.pdf in literature/.*

> *Rework paragraph 3 of the introduction in paper.tex.*

> *Check the equation in line 142 of paper.tex.*

- **Output:** have it write straight into the final file. It finds the right place.

> *Slide 17 is fine as discussed. Write it into slides.tex.*

- Never copy answers from the chat window into your documents.

Script everything

- Every table, figure and number in your document comes from a script. Never edit generated tables by hand: you will redo each one ten more times, so make it come out right from the code.
- You don't need to know how to script an operation: the agent does it for you.

code/wage_regs.do writes the table:

```
* Example data shipped with Stata
sysuse nlsw88, clear
generate lwage = ln(wage)

* Regressions
eststo clear
eststo: regress lwage grade ttl_exp, robust
eststo: regress lwage grade ttl_exp union, robust

* Export a production-ready LaTeX table
esttab using "output/tables/wage_regs.tex", ///
    replace booktabs label se               ///
    keep(grade ttl_exp union)               ///
    mtitles("Baseline" "With union")
```

paper.tex includes it:

```
\begin{table}
  \caption{Wage regressions}
  \input{output/tables/wage_regs.tex}
\end{table}
```

✗ Don't	✓ Do
Change the column order of Table 2 in paper.tex.	Change wage_regs.do so that Table 2 comes out in this column order.
What does this coefficient imply in euros? (Then paste the answer.)	Add this calculation to the do-file, export the number, and use it in Section 4.

instructions.md: the contract

Section	My typical content
Scope	We are writing a research paper in labour economics.
Literature	Read the papers in literature/.
Stages	sub-steps, and how we proceed and interact (longer projects)
Output	The main document is paper.tex.
Software	which software; where code, input and output files are
Style	a writing sample of mine in samples/: sets layout, language and tone
Checks	At the end: grammar, language, style and consistency check.
Logging	Record progress and TODOs in status.md; read it at the start of each session.

- Instead of long one-off prompts: one file that sets the scope and the rules.
- **Format barely matters:** no syntax rules, .md or .txt. A little structure (headings) helps.
- **Why bother:** 10–15 minutes of writing forces you to think through background, deliverables and sub-steps, and tells your RA clearly what to do.
- Extend it as you go.

Example: the `instructions.md` behind these slides

```
# Instructions

## Scope
We are writing an overview on jump-starting AI agents for economists. The main goal
        is to help other economists get started as quickly as possible: only the
        bare-bones essentials to keep in mind, with no unnecessary detail,
        background information or technical lingo. Readers should then be able to
        start working on a project of their own right away.

Caveat to convey throughout: this is my personal take in a very rapidly changing
        field. The workflow and approach are what work for me, and I'm happy to
        discuss them with anyone at any time.

## Background material
- `background.md`: my collection of ideas and thoughts I'd like to convey. They are
        not yet well structured.
- `recordings/`: the raw voice transcripts behind parts of `background.md`. Use them
         for reference only; `background.md` is the cleaned-up version.

## Tasks and stages
1. **Structure:** read `background.md` and propose how to bring the content into a
        proper structure: categories, sections and sub-bullets, in a sensible order
        . Do not write the presentation yet.
2. **Deliberate:** we walk through the proposed structure together and agree on the
        basic layout and on what should be conveyed. Where useful, suggest:
   - additional details or points I may have missed
   - concrete examples, e.g. short code snippets, an example `instructions.md`, or
        an example file/directory structure for a project folder
   - content that could be dropped or merged because it is not essential for a quick
         start
3. **Presentation:** once the structure is agreed, produce the slides (see below).

Before starting a stage, plan the subtasks and ask me about anything unclear up
        front. Anything non-critical can be left as a TODO note for later.

Whenever anything is unclear, ask me via AskUserQuestion with options I can click on
        , rather than writing a list of questions in the chat.

## Expected output
- A LaTeX Beamer presentation, `slides.tex`, compiled to `slides.pdf` at the end.
- Minimal layout. Where useful, use a few icons, light blue colouring, and
        structuring elements such as two-column/table layouts. Make a suggestion to
        me before committing to a design.
- Keep all text in the `.tex` source. Any figures, tables or code examples should be
         generated or included from files, not pasted in by hand.

## Style
- Plain, concise and practical, aimed at economists rather than programmers.
- Bullet-point style suitable for slides. No hype.

## Checks
At the end (or whenever I ask): a thorough grammar, language, style and internal
        consistency check of the slides, including consistency with `background.md
        `.

## Status
At the end of each session, after major changes, or whenever I ask, record what we'
        ve done and the open TODOs in `status.md`. At the start of each session,
        read `status.md` first.
```

Ways of working

Asynchronous

- Longer tasks run in the background: half an hour, a few hours, overnight. Run several agents on different projects; reconcile when they finish.
- Settle all decisions up front, or defer non-critical ones as TODOs. Avoid: the agent stops after an hour with a question.

> *Plan the subtasks for this stage and ask me about anything unclear before you start. Leave non-critical points as TODOs.*

Sparring partner

- Test ideas before taking them to colleagues: does the argument hold up against what we wrote elsewhere, and against the literature?
- Often points out what you have overlooked, such as a contradiction with an earlier section.

> *I think we should reframe the discussion in Section 5 as follows: . . . Assess this critically against the rest of the paper and the literature in literature/.*

Working interactively on one part is fine too, but most tasks can run asynchronously.

Pitfalls

- **Made-up or outdated citations.** My rule in `instructions.md`:

> *NEVER cite (add to the .bib file) any document that is not in literature/.*

  The agent may suggest literature, but I look up and download the papers myself. This way I have the real, latest versions, and cite only what I chose.
- **Backups.** The agent writes into your files. Keep file history (Dropbox, OneDrive, . . .), and read permission prompts before approving them.
- **Confidential data.** Check data licences and your institution's rules before giving the agent access to confidential data.

Putting it together: the jump start

1. Create a folder for the project.
2. Put in the background files: literature, data, a writing sample, your draft or notes.
3. Write `instructions.md` (10–15 minutes).
4. Open the agent in that folder (e.g. VS Code with Claude Code).
5. Tell it to read the instructions, and take it from there.

First prompts

> *Read instructions.md. Plan the subtasks and ask me about anything unclear before you start.*

> *Read status.md and continue where we left off.*

> *Do a thorough grammar, style and consistency check of paper.tex.*

Questions, or a workflow that works better for you? Happy to discuss: simon.martin@univie.ac.at